Dhruv Prajapati

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4th year Mechanical Engineering student, passionate about **thermal/mechanical** roles, seeking a **spring/summer 2027 internship**

Education

Rochester Institute of Technology **Aug. 2023 – Dec. 2027**
Bachelor of Science in Mechanical Engineering, Senior Year, Dean's List 2026, GPA 3.50 Rochester, NY

Experience

Battery Cell Engineer Intern **Sept. 2026 – Dec. 2026**
TESLA Austin, TX

- **Tabless 4680 Winder** | *TwinCAT, Electro-mechanical sensors, DOE*
- Optimized the speed process that reduced jelly-roll winding time by 200 ms, increasing throughput 33.3% and delivering 9.5 million extra cells annually fleet-wide - enabling higher output of Tesla Semi trucks, Cybertrucks, and related vehicles.
- Led a transverse-direction (TD) alignment improvement on a high-volume battery cell winding line, cutting yield loss by 20%.
- Analyzed 13+ winder machine signals using TwinCAT Scope View to isolate tension and alignment variations, optimizing electro-pneumatic controllers (EPCs) and servo data to improve battery jelly-roll winding quality.

Failure Analysis Engineer Intern **Jun. 2026 – Aug. 2026**
PG&E, Pacific Gas & Electric Company San Ramon, CA

- **External Aerodynamics & Multiphysics Analysis** | *ANSYS Fluent + Mechanical + Maxwell, MATLAB*
- Developed a transient external-aerodynamics CFD and FSI framework to evaluate wind loading on a 230-kV porcelain post insulator, coupling ANSYS Fluent pressure fields with structural FEA.
- Characterized vortex shedding across 27–54 m/s crossflow using FFT and Strouhal scaling; transferred CFD pressure fields through one-way FSI, predicting a factor of safety > 23.
- Built a 3D external-flow CFD model using BOI mesh refinement and SST $k-\omega$ turbulence modeling; evaluated pressure and drag across 27–54 m/s crossflow and validated aerodynamic trends against analytical predictions.

Mechanical Test Engineer Intern **Aug. 2025 – Dec. 2025**
MKS Instruments Rochester, NY

- **Liquid-cooled Heat Sink Analysis** | *SolidWorks Flow Simulation (CFD), Experimental test bench*
- Performed CFD and experimental characterization of a liquid-cooled microchannel heat sink for a 5-kW pulsed-DC generator, evaluating pressure-drop and flow-rate tradeoffs across operating conditions.
- Validated simulated pressure-drop predictions against experimental measurements within $\pm 2\sigma$; identified flow maldistribution and design improvements that reduced hydraulic pressure drop by 25%.
- Developed first-principles fluid-flow models in MATLAB to diagnose flow maldistribution across R&D test benches, correlating flow-sensor and pressure-gauge measurements with pump performance curves.

Fluids & Heat Transfer Research **Feb. 2026 – May 2026**
FAST LAB, Fluids Acoustics Structures Laboratory Rochester, NY

- **Vortex-Ring Fluid Dynamics & Experimental Validation** | *Arduino, Pololu Jrk Controller, 3D Printing*
- Conducted PIV experiments to characterize vortex-ring flow structures, propagation, and impingement behavior in confined geometries for fluid-flow and heat-transfer research.
- Designed and built a repeatable vortex-ring test platform integrating a linear actuator, Arduino, Pololu motor controller, and 3D-printed fixtures, reducing experimental setup/execution time by 35%.

Technical Skills

Aerodynamics & CFD: External Aerodynamics, Transient CFD, FSI, Vortex Shedding

Thermal & Fluid Systems: Heat Transfer, Fluid Mechanics, CFD (SolidWorks Flow Simulation, ANSYS Fluent)

CAD Design & Analysis: SolidWorks, ANSYS Mechanical & Fluent, CATIA v5

Programming: Python, MATLAB, Simulink, LabVIEW, Beckhoff TwinCAT 3

Hardware & Testing: Hitachi SEM, Oscilloscopes, Keysight Waveform generator, PIV, Instron

Manufacturing: CNC Milling, Lathe, GD&T, 3D Printing, Surface Metrology